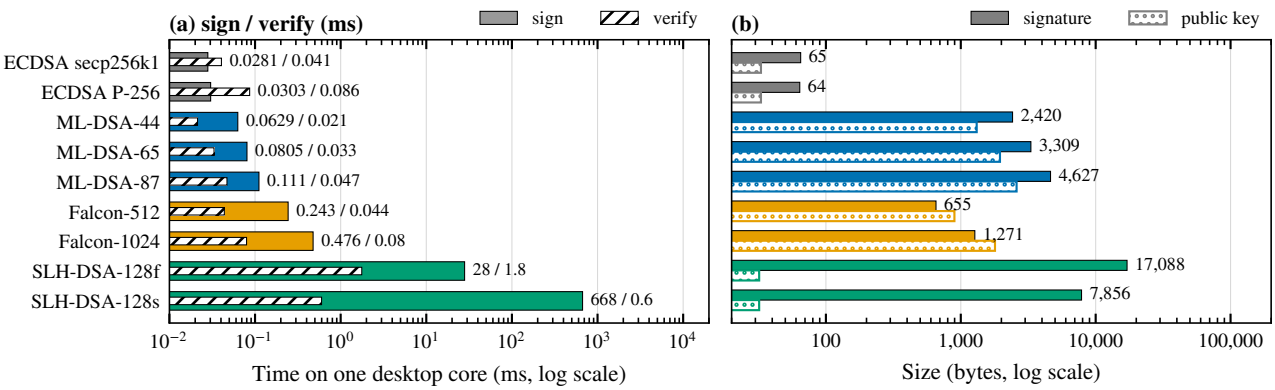

(a) sign / verify (ms)
sign
verify
ECDSA secp256k1
ECDSA P-256
ML-DSA-44
ML-DSA-65
ML-DSA-87
Falcon-512
Falcon-1024
SLH-DSA-128f
SLH-DSA-128s
0.0281 / 0.041
0.0303 / 0.086
0.0629 / 0.021
0.0805 / 0.033
0.111 / 0.047
0.243 / 0.044
0.476 / 0.08
28 / 1.8
668 / 0.6
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Time on one desktop core (ms, log scale)
(b)
signature
public key
65
64
2,420
3,309
4,627
655
1,271
17,088
7,856
100
1,000
10,000
100,000
Size (bytes, log scale)